

Astro Night Planner 1.4.0

Practical manual for planning astrophotography nights with weather, Moon, object selection, equipment, horizon, framing and local surveys

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Overview and operating concept

Astro Night Planner is a practical planning tool for astrophotography nights. It combines visibility, twilight, Moon, weather, personal equipment, location, horizon and framing in one workflow.

Planning

Date, location, planning window, weather, object filters, object list, object details, altitude curve, horizon view and Aladin framing.

Settings

Equipment, locations, horizon profiles, display profiles, weather and quality profiles, local surveys, backup and help.

Data storage and backup

Profiles, equipment, locations, horizon profiles and settings are stored locally in the browser. IndexedDB holds the user data; Cache Storage holds the PWA app files. Cookies are not the main storage.

Create regular backups: Deleting site data, using another browser profile or changing the domain can lead to a separate or empty data set. Keep external backups outside the browser.

Planning night, date buttons and Moon

The date buttons summarize the selected night. Weather and Moon information refer to the selected planning window, not automatically to the full sunset-to-sunrise night.

Display	Meaning
↑ Moonrise	Moonrise time if it lies in or near the relevant night window.
▲ Moon altitude	Maximum Moon altitude during the selected planning window.
↓ Moonset	Moonset time. If it lies outside the planning window, the planning data states this explicitly.

The planning data may also show Moon culmination for the full night. If that culmination lies before or after the selected planning window, the app adds a clear note. This separates maximum Moon altitude in the planning window from the whole-night culmination.

Object filters and object list

Start with broad filters and narrow the list afterwards. Minimum altitude, minimum duration, Moon distance, object types, catalogs, capture filters, magnitude and object size work together. Direct search finds objects independently of the other filters.

The object list can be adjusted through display profiles. Columns can be shown, hidden and reordered. Click an object row to open the detail view.

Equipment, reducer and Aladin framing

Telescope, reducer/corrector/Barlow and camera determine effective focal length, field of view, pixel scale and the camera frame in Aladin.

Example: A 700 mm telescope with a 0.80x reducer gives 560 mm effective focal length. With the same camera, the field of view becomes larger; with a 1.50x Barlow it becomes smaller.

In the Aladin view, telescope, reducer/corrector/Barlow and camera can be combined temporarily. If the combination is not saved as a setup, the selection does not jump back. The app calculates the framing temporarily and shows a notice. Use **Save as setup** to store the combination.

Local surveys and Local Survey Server

Local surveys are local copies of HiPS sky maps. They speed up display, reduce internet dependency and allow large surveys to remain available even without an internet connection. The planner uses local

surveys only when they are enabled, reachable and correctly configured.

Why a local web server is required

A browser app cannot read a normal Windows folder such as `D:\AstroSurveys\nsns\ohs8` directly. The folder must be served through a local HTTP address. The recommended survey file-server address is `http://127.0.0.1:8765/`. This address points to your own computer and is not a public internet server.

Windows helper

The **ANP Local Survey Server 1.0** package contains a single EXE. After unzipping, start `ANP-Local-Survey-Server-1.0.exe` by double-clicking it. A small control window and the browser-based configuration interface open.

Control	Function
Open interface	Opens the local configuration page. Use it if the browser did not open automatically.
Start server	Starts the local survey file server. The HiPS files are reachable through the base URL only after this server is running.
Stop server	Stops only the survey file server. The helper and configuration interface remain available.
Exit	Stops the survey server, stops the configuration interface and exits the helper completely. Use this before replacing or deleting the EXE.
X in the control window	Hides the window to the notification area. It does not exit the helper.

Tray menu, autostart and background mode

The tray icon is in the Windows notification area. Right-click it to open a menu with **Open interface**, **Start server**, **Stop server** and **Exit**. The configuration interface offers these options:

- **Start with Windows:** creates an autostart entry for the current Windows user.
- **Start in background / tray:** starts without a visible browser window and remains available through the tray icon.
- **Automatically start local survey server:** starts the file server as soon as the helper is running.

Console mode is available with `ANP-Local-Survey-Server-1.0.exe --console`. Background mode uses `--background`.

Can the planner start the Windows program?

The planner cannot directly start the Windows program. Browsers block this for security reasons. If the helper is already running, the **Open Local Survey Server** button in the survey settings can open the local configuration interface. If it is not reachable, start the EXE manually or enable Windows autostart.

Recommended folder structure

Use one common root folder, for example `D:\AstroSurveys`. Survey folders are placed below it. The short codes come from the survey addresses and are entered in the planner as relative paths.

Local folder	Relative path	Description
<code>D:\AstroSurveys\nsns\ohs8</code>	<code>nsns/ohs8/</code>	NSNS OIII/H-alpha/SII composite
<code>D:\AstroSurveys\nsns\halpha8</code>	<code>nsns/halpha8/</code>	NSNS H-alpha
<code>D:\AstroSurveys\nsns\oiii8</code>	<code>nsns/oiii8/</code>	NSNS OIII

Local folder	Relative path	Description
D:\AstroSurveys\nsns\sii8	nsns/ sii8/	NSNS SII
D:\AstroSurveys\nsns\hbr8	nsns/ hbr8/	NSNS H-beta/ red or related product if available
D:\AstroSurveys\nsns\rgb8	nsns/ rgb8/	NSNS RGB product if available

Content of a HiPS folder

A HiPS folder must contain at least a readable `properties` file. For display, the `Norder...` tile folders are also required, for example `Norder3`, `Norder4` and higher orders. Depending on the survey, `Moc.fits` and `Allsky...` files are present. The `properties` file contains `hips_tile_format`. If it says `png`, PNG tiles are used; the app must not force JPEG.

Configuration in the planner

1. Open **Settings - Central settings - Aladin - Survey - Local survey sources**.
2. Enable **Enable local survey sources**.
3. Enter the base URL `http://127.0.0.1:8765/`.
4. Keep **Prefer local sources when available** enabled.
5. Decide whether **Use online source when local source is unavailable** should be enabled. With fallback enabled, the online source is used when local data is unavailable. Without fallback, an error is shown.
6. In each survey row, enable **Prefer local source** and enter the relative path, for example `nsns/ohs8/`.
7. Use **Check local survey source**. The check verifies the local URL and the HiPS `properties` file.

Fallback logic and diagnostics

When the sky image is opened, the app checks the local source first if local use and local preference are enabled. If it is reachable, the local URL is passed to Aladin. If it is not reachable and online fallback is allowed, the app uses the online source. If fallback is disabled, the app shows an error instead of silently switching to DSS or another default background.

Example: Root folder `D:\AstroSurveys` , local folder `D:\AstroSurveys\nsns\ohs8` , base URL `http://127.0.0.1:8765/` , relative path `nsns/ohs8/` . The resulting URL is `http://127.0.0.1:8765/nsns/ohs8/` . The file `http://127.0.0.1:8765/nsns/ohs8/properties` must be readable in the browser.

Weather, cloud model and external sources

The model consensus combines several forecast models. The app's astro cloud map helps evaluate cloud movement, model deviation and precipitation. Meteoblue, Clear Outside, Windy and Ventusky are external reference sources; they load separately and have their own timelines.

Location, horizon and N.I.N.A.

Locations contain name, coordinates, elevation and time zone. Horizon profiles describe local obstructions by azimuth and altitude. They affect visibility, horizon curve and planning. N.I.N.A. horizons can be imported and exported.

Update, PWA and troubleshooting

The installable web app uses a service worker and Cache Storage. When local survey requests are used, local hosts such as

127.0.0.1 and localhost are not replaced by the app cache; they are passed directly to the local server.

- If local surveys are not visible: check that the Local Survey Server is running and that /properties is readable in the browser.
- If the helper cannot be replaced: choose **Exit** in the control window or tray menu, not only X.
- If the configuration page does not open automatically: use **Open interface** in the control window or tray menu.

Glossary

Term	Explanation
HiPS	Hierarchical Progressive Survey, a tiled sky-map format.
properties	Control file of a HiPS survey with information such as tile format and order.
Norder	HiPS tile-folder structure by order/resolution level.
Fallback	Switching to the online source when the local source is unavailable and fallback is allowed.
Tray	Windows notification area at the right side of the taskbar.

Astro Night Planner 1.4.0 - User manual. This manual describes how to use the app and does not replace checking actual local conditions.